

NEAR-INFRARED *H*-BAND FEATURES IN LATE O AND B STARS

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ABSTRACT

We examine the spectral characteristics of normal OB stars with high-signal-to-noise ratio (>120) *H*-band ($1.6\ \mu\text{m}$) spectra at a resolution of 2000. We find that several atomic lines vary smoothly with stellar temperature, as first shown by Blum et al. However, we find a previously unreported, significant variation in the strength of some of these lines with stellar luminosity. B supergiant stars show stronger He I and weaker Br 11 as compared with low-luminosity B dwarf stars of the same spectral class. It is for this reason that luminosity class must also be determined to obtain an accurate spectral type for a given star using *H*-band spectra. We suggest a method for estimating the spectral type and luminosity of an OB star over the wavelength range from 1.66 to $1.72\ \mu\text{m}$ using hydrogen Br 11 at $1.681\ \mu\text{m}$, He I at $1.700\ \mu\text{m}$, and He II at $1.693\ \mu\text{m}$. The use of the near-infrared spectral range for classification has obvious advantages over optical classification when applied to heavily reddened stars, such as in star-forming regions or deeply embedded lines of sight within the plane of the Galaxy, such as the Galactic center. Furthermore, the *H* band is less likely to be contaminated by infrared excess emission, which is frequently seen around massive young stellar objects beyond $2\ \mu\text{m}$.

Key words: atlases — infrared radiation — stars: early-type — stars: fundamental parameters

1. INTRODUCTION

High-quality, near-infrared spectra (from 1.0 to $2.4\ \mu\text{m}$) of numerous stars were first presented by Johnson & Mendez (1970) nearly 30 years ago. However, use of near-infrared spectra, particularly for spectral classification, has become popular only in the last few years (Ali et al. 1995; Greene & Meyer 1995; Hanson, Conti, & Rieke 1996, hereafter HCR; Greene & Lada 1996; Luhman & Rieke 1998; Meyer et al. 1998). The ease with which one can now obtain resolutions commensurate with those used for optical classification, even for moderately faint stars, has been important in the practical use of near-infrared spectroscopy for classification work. The great utility of near-infrared classification is its ability to determine fundamental properties of stars lying behind large amounts of extinction, where traditional optical classification is impossible. When the visual extinction, A_V , becomes larger than about 7 or 8 mag, near-infrared spectral classification becomes essential.

The use of the strong He I $1.700\ \mu\text{m}$, $4p\ ^3P^o - 3s\ ^3S$, absorption line as a temperature diagnostic in OB stars was first introduced by Blum et al. (1997). In their initial *H*-band study of nine stars obtained at a resolution of 570, Blum et al. showed a smooth variation of $1.700\ \mu\text{m}$ He I line strength with spectral type. While the function is double-valued, their sample suggested that mid O and mid and late B stars, both of which have weak He I, could be differentiated by the strength of their hydrogen features (mid and late B stars having much stronger hydrogen lines). Unfortunately, this initial study included too few stars to constrain the behavior of He I adequately as a function of both spectral type and luminosity class. With this in mind, we have obtained new near-infrared *H*-band spectra with several times the signal-to-noise ratio (S/N), resolution, and sample size of the Blum et al. survey. Our goal is to more accurately constrain the

behavior of the He I line as a function of stellar characteristics (temperature and luminosity) and to explore the utility of the *H*-band spectral region for the classification of heavily reddened OB stars.

2. *H*-BAND SPECTRA

Our sample of 34 spectroscopic standard stars is given in Table 1. An important aim of our study is to look for variations in line strength with luminosity within the same spectral class. We have chosen a set of both dwarf and supergiant late O through late B stars, neglecting giant stars in this initial study.

2.1. Observations

All observations were made using the Steward Observatory near-infrared long-slit spectrometer, FSPEC (Williams et al. 1993), which uses a 256×256 NICMOS3 array. Spectra were taken using the Multiple Mirror Telescope (MMT) and the Bok 2.3 m Telescope in 1997 June (MMT), October (MMT), November (Bok), and December (Bok). Spectra were taken with the high-resolution, $600\ \text{line mm}^{-1}$ grating, yielding a 2.5 pixel resolution element of $8.7\ \text{\AA}$, or $R \approx 2000$ at $1.70\ \mu\text{m}$. The spectrometer has a slit size of $1'2 \times 32''$ on the MMT, allowing for four unique stellar positions along the slit. On the Bok Telescope, the slit is $2'4 \times 90''$, allowing for six unique positions in the slit. The same observing and reduction procedures were used for all stars observed at both telescopes.

The sample was chosen to fully represent late O and B spectral types. However, only dwarf and supergiant luminosity classes were included, to search for luminosity effects in the line strengths. When possible, the stars were chosen from the HCR *K*-band spectral atlas, though new stars have been added to complete our spectral or luminosity coverage when suitable stars were not available from the HCR atlas in the northern sky.

Each target star was observed in several unique positions along the slit. The typical integration times for the standard stars were between 5 and 20 s per slit position, giving total integration times of 30 s to a few minutes. After dark current

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TABLE 1
STARS IN *H*-BAND SPECTRAL ATLAS

Star	Optical Spectral Type	Reference	S/N	Date Observed	Notes
HD 14542	B8 Ia	1	220	1997 Jun	
HD 14956	B1.5 Ia	1	250	1997 Jun	
HD 15497	B6 Ia	1	200	1997 Jun	
HD 21071	B7 V	2	200	1997 Dec	
HD 21291	B9 Ia	1	170	1997 Nov	
HD 28375	B3/B5 V	2	120	1997 Nov	
HD 31726	B1 V	3	100	1997 Nov	
HD 34762	B9 V	4	150	1997 Dec	
HD 36371	B4 Ia	1	150	1997 Dec	
HD 36959	B1.5 V	5	140	1997 Nov	NGC 1977 482
HD 36960	B0.5 V	5	140	1997 Nov	NGC 1977 493
HD 37128	B0 Ia	5	140	1997 Nov	46 Ori
HD 37468	O9.5 V	6	140	1997 Dec	σ Ori
HD 38771	B0.5 Ia	5	100	1997 Dec	κ Ori
HD 47839	O7 V((f))	7	100	1997 Dec	15 Mon
HD 48279	O8 V	8	120	1997 Dec	
HD 169454	B1 Ia ⁺	5	150	1997 Oct	
HD 169990	B8 V	4	170	1997 Oct	
HD 183143	B7 Ia	3	150	1997 Oct	HT Sge
HD 188001	O7.5 Iaf	7	140	1997 Oct	9 Sge
HD 190603	B1.5 Ia ⁺	1	150	1997 Oct	
HD 193237	B2p (LBV)	4	120	1997 Oct	P Cyg
HD 193322	O9 V:(n))	7	110	1997 Oct	
HD 193514	O7 Ib(f)	7	150	1997 Oct	
HD 194279	B1.5 Ia	1	170	1997 Oct	NGC 6910 2
HD 194280	OC9.7 Iab	7	120	1997 Oct	NGC 6913 105
HD 195592	O9.7 Ia	7	150	1997 Jun	
HD 198183	B5/B6 V	2	150	1997 Oct	λ Cyg
HD 198478	B2.5 Ia	1	170	1997 Jun	55 Cyg
HD 198784	B2 V	4	150	1997 Oct	
HD 210809	O9 Iab	8	180	1997 Jun	
HD 224055	B3 Ia	1	220	1997 Jun	
CygOB 2 12	B5 Ia	9	180	1997 Jun	
CygOB 2 MT138	O8.5 I	9	180	1997 Jun	

REFERENCES.—(1) Lennon, Dufton, & Fitzsimmons 1992; (2) Hoffleit & Jaschek 1982; (3) Morgan, Code, & Whitford 1955; (4) SIMBAD database; (5) Walborn & Fitzpatrick 1990; (6) Conti & Alschuler 1971; (7) Walborn 1972; (8) Walborn 1973; (9) Massey & Thompson 1991.

subtraction and flat-fielding, sky emission and additional thermal background were removed by subtracting one slit position from the next. Next, the two-dimensional spectra were integrated along the spatial direction of the slit and wavelength-calibrated using the OH sky emission lines (Steed & Baker 1979).

Interspersed between the observations of our target stars we obtained spectra of A dwarf and solar analog telluric standard stars, used to remove the atmospheric absorption features. A near-infrared spectrum of the sun was first divided out of the solar analog star's spectrum (between G1.5 and G3 V is required for the solar analog star at this rather high resolution and S/N). This spectrum, now representing the atmospheric transmission, was then divided out of the raw spectrum of the A dwarf star. The remaining A dwarf spectrum exhibited only the intrinsic hydrogen lines, which we directly fit. The solar analog standard star was used only to remove telluric absorption features in the region of the A dwarf hydrogen lines. With a direct fit of the A star hydrogen lines, we returned to the raw spectrum of the A star and removed the hydrogen lines with the previously determined fit. The remaining spectral features seen at our resolution and S/N in the A dwarf spectrum are due solely to atmospheric absorption lines. The A dwarf spectrum, now free of hydrogen lines, was used to remove telluric contamination from the target O and B stars (see the appendix to HCR for an illustrated description of this

method). Typically, a separate A telluric standard star was observed to within one or two hundredths of an air mass from each OB star.

2.2. Line Identifications from 1.66 to 1.72 μ m in OB Stars

The standard star spectra are shown in Figures 1, 2, and 3. The *H*-band spectrum of P Cygni is found in Figure 4. The wavelength coverage allowed us to measure the 1.700 μ m He I (4–3) line simultaneously with the 1.681 μ m Brackett H I (11–4) transition. Between these two stronger lines lies the He II (12–7) line at 1.693 μ m. Typically, we were able to achieve S/Ns in excess of 120. For the He I and He II lines, the errors are rather small, as we are fairly confident of the telluric corrections in those spectral regions. We can identify features as small as 0.15 to 0.20 Å at these positions, depending on the S/N. Despite the high S/N, the Br 11 feature cannot be measured with such confidence. Systematic errors dominate in the correction of the hydrogen line in the telluric standard stars, which can be quite large (see § 2.1). Furthermore, very broad features, such as those seen in the hydrogen lines of the dwarf stars, are difficult to measure accurately because of difficulty in continuum placement. The equivalent width measurements should not be trusted to better than 0.3 Å for the Br 11 lines, and 0.5 Å is a more typical error. Equivalent width measurements and their errors for all standard stars are given in Tables 2 and 3.

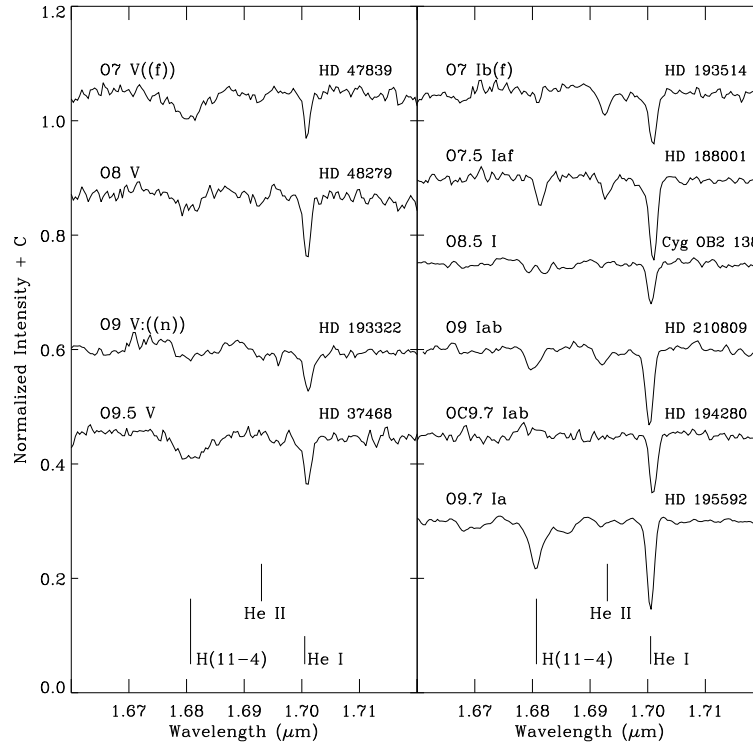


FIG. 1.—*H*-band O star standards. Features identified in the figure include Brackett (11–4) at 1.6814 μm , He II at 1.693 μm , and the 1.700 μm He I line. Resolution is 2000, S/N varies from 120 to 180.

Upon inspection of Figures 1, 2, and 3, and using Figure 5, which graphically presents the line strengths versus spectral type, we can examine the behavior of the stellar features with spectral type and luminosity class. We chose to include

P Cygni in this study to demonstrate the dramatic differences in its spectral characteristics at near-infrared wavelengths compared with normal B supergiants (Fig. 4). We did not include the measurements made of P Cygni in

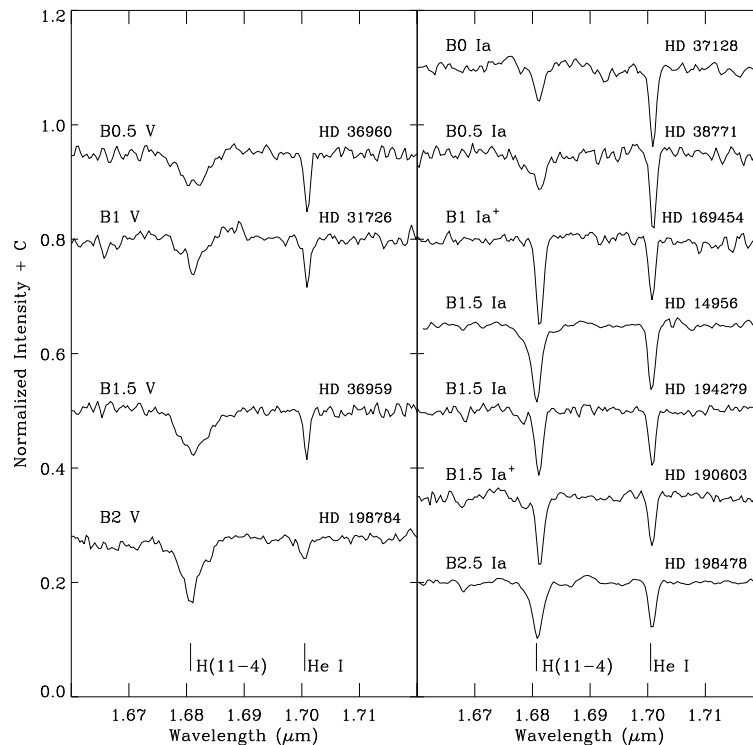


FIG. 2.—*H*-band early B star standards. Features identified in the figure include Brackett (11–4) at 1.6814 μm and the 1.700 μm He I line. Resolution is 2000, S/N varies from 100 to more than 200. P Cygni, the luminous blue variable supergiant with early B spectral type, is shown separately in Fig. 4.

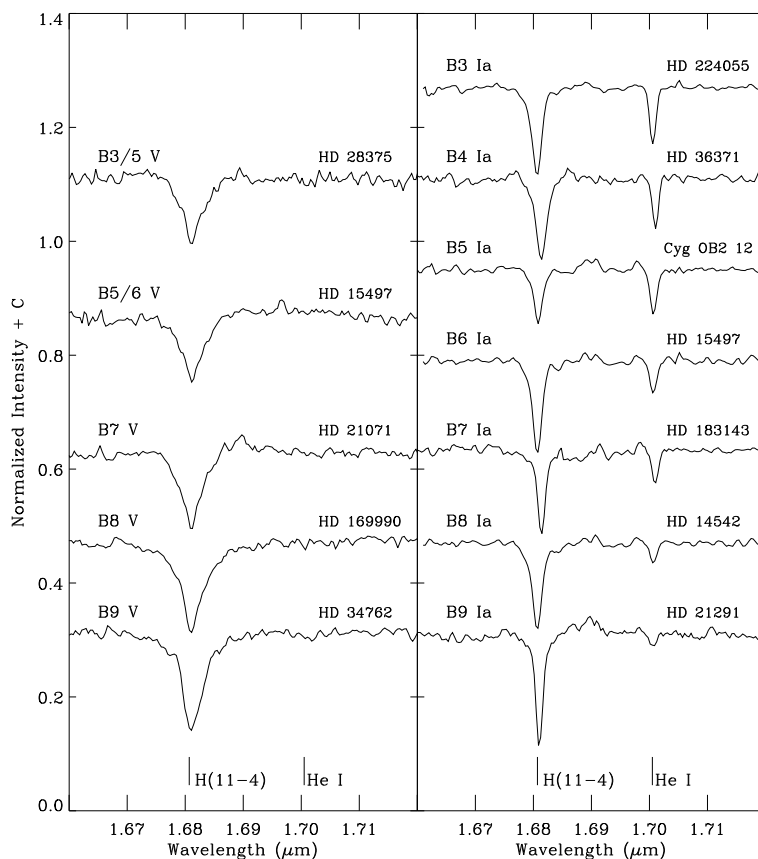


FIG. 3.—*H*-band mid and late B star standards. Features identified in the figure include Brackett (11–4) at 1.6814 μm and He I 1.700 μm . Resolution is 2000, S/N varies from 120 to 200.

Figure 5. P Cygni is a highly luminous, variable early B supergiant. A more thorough study that also includes high-quality *J*- and *K*-band spectra of P Cygni is in progress (Voors & Hanson 1998).

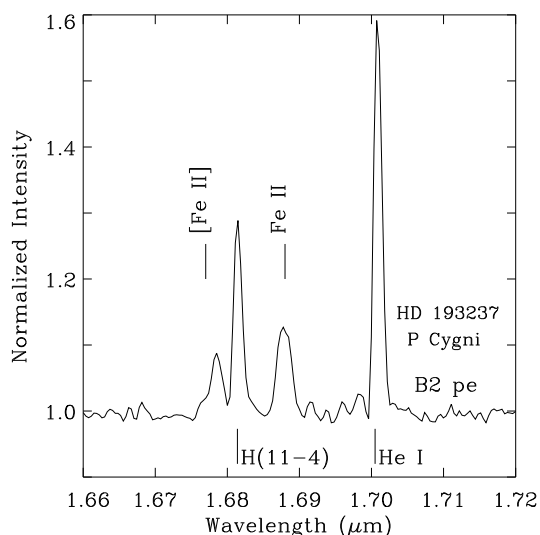


FIG. 4.—*H*-band spectrum of the luminous blue variable star HD 193237 (P Cygni). P Cygni is an extremely luminous early B star. Note that Brackett (11–4) at 1.6814 μm and the 1.700 μm He I line are seen in emission, distinctively different from the early B supergiants shown in Fig. 2. P Cygni also shows emission lines of [Fe II] at 1.677 μm and Fe II at 1.688 μm . This is typical of highly evolved, massive supergiants, which are in a short-lived and thus rare evolutionary state (see Morris et al. 1996).

As expected, the (12–7) 1.693 μm He II line (Fig. 1) is detected only in the O stars. Though it is weak, further study of this line's behavior in early and mid O stars should prove useful for identifying O stars suffering from significant near-infrared excess at longer wavelengths. For instance, the (10–7) transition of He II at 2.1885 μm is particularly useful for classification of O stars (HCR), but this spectral region is known to suffer thermal contamination from warm circumstellar material in very young stars even of relatively high mass (Hanson, Howarth, & Conti 1997; Testi et al. 1998). Interestingly, the He II line at 1.693 μm appears to be stronger in late O supergiants than late O dwarfs of approximately the same spectral type (Figs. 1 and 5). While it is not mentioned by the authors, a similar effect can be seen in the behavior of He II at 2.1885 μm in late O supergiants and dwarfs in the HCR *K*-band atlas. Early model atmospheres predicted such a variation with gravity for the He II 5411 Å (Auer & Mihalas 1972), namely, stronger equivalent width in lower gravity late O stars. This possible luminosity effect in the near-infrared He II lines needs to be explored further with a spectroscopic study that includes more early and mid O stars and comparisons with model predictions.

We identify three varying effects occurring in the Br 11 feature among our sample of standard stars. Cooler OB stars have stronger Br 11 absorption, and dwarf stars show broader Br 11 features than the supergiants. These effects were also noted by Blum et al. (1997) and Meyer et al. (1998). We also note that the slope of the Br 11 equivalent width versus spectral type seen in our study is consistent

TABLE 2
EQUIVALENT WIDTHS (Å) FOR O STARS IN SPECTRAL ATLAS

Star	Optical Spectral Type	Br 11 1.6814 μm	He II 1.693 μm	He I 1.7004 μm
HD 47839	O7 V((f))	3.0 ± 0.5	0.3 ± 0.2	1.0 ± 0.2
HD 193514	O7 Ib(f)	<1.0	0.8 ± 0.2	1.6 ± 0.2
HD 188001	O7.5 Iaf	0.8 ± 0.2	0.7 ± 0.2	2.5 ± 0.2
HD 48279	O8 V	1.6 ± 0.5	0.5 ± 0.2	1.6 ± 0.2
Cyg OB2 MT 138	O8.5 I	^a	...	1.1 ± 0.2
HD 193322 ^b	O9 V: ((n))	<1.0	...	1.1 ± 0.2
HD 210809	O9 Iab	1.0 ± 0.3	0.5 ± 0.2	2.2 ± 0.2
HD 37468	O9.5 V	2.3 ± 0.5	...	1.4 ± 0.2
HD 194280	OC9.7 Iab	<1.0	...	1.6 ± 0.15
HD 195592	O9.7 Ia	2.4 ± 0.3	...	2.3 ± 0.2

^a Emission plus absorption?

^b Br 11 and Br 12 appear to be absent in the spectrum.

with those obtained by others using similar resolution *H*-band spectroscopy (M. Meyer 1998, private communication).

3. SPECTRAL CHARACTERISTICS OF OB STARS IN THE *H*-BAND

The variations in spectral characteristics seen in Figures 1, 2, and 3 suggest that the *H* band may serve as a useful spectroscopic and luminosity indicator for heavily extinguished OB stars, just as the *K* band has proved to be (HCR). In O stars, optical spectral type is mostly strongly based on the ratio between neutral and ionized helium. The same criterion can be applied at near-infrared wavelengths, specifically, the ratio of 2.1126 μm He I line to 2.1885 μm He II line is used by HCR. Unfortunately, the 2.1126 μm He I line can be contaminated by a neighboring singlet He I transition at 2.1134 μm and by N III at 2.1155 μm (HCR). The neutral helium line at 1.7004 μm is a stronger singlet transition than the He I line in the *K* band, and is also free of any nearby contaminating lines. The He II (12–7)

transition at 1.693 μm is slightly weaker than the *K*-band He II (10–7) transition. However, just as in the *K* band, the *H*-band He II line is detected in O stars out to O9 in our sample. Thus, the *H*-band He II line remains a reliable identifier for an O star atmosphere.

There may be several benefits to using the *H*-band classification lines over the *K*-band lines in B stars. For one, near-infrared veiling or excess emission, seen to exist around very young stars and in some evolved massive stars, is not significant at *H*-band wavelengths. Typical near-infrared veiling is caused by thermal dust emission, occurring in the circumstellar material of these objects. Typical dust destruction temperatures of $T \approx 1500$ K mean there will be little thermal emission at wavelengths as short as 1.7 μm . Furthermore, the three main lines, the Br 11, He II, and He I transitions, lie in a compact spectral region, making them easy to observe with limited spectral coverage. Furthermore, if veiling is occurring and the emission has a flux dependence with it, it will not strongly affect the relative strengths of the *H*-band classification lines, on account

TABLE 3
EQUIVALENT WIDTHS FOR B STARS IN SPECTRAL ATLAS

Star	Optical Spectral Type	Br 11 1.6814 μm (Å)	He I 1.7004 μm (Å)
HD 37128	B0 Ia	1.5 ± 0.5	1.9 ± 0.2
HD 36960	B0.5 V	2.8 ± 0.5	1.2 ± 0.2
HD 38771	B0.5 Ia	2.3 ± 0.5	1.9 ± 0.2
HD 31726	B1 V	2.5 ± 0.5	1.0 ± 0.2
HD 169454	B1 Ia ⁺	2.4 ± 0.3	1.3 ± 0.15
HD 36959	B1.5 V	4.5 ± 0.5	1.0 ± 0.2
HD 14956	B1.5 Ia	3.2 ± 0.3	1.7 ± 0.15
HD 194279	B1.5 Ia	1.8 ± 0.3	1.3 ± 0.15
HD 190603	B1.5 Ia ⁺	1.9 ± 0.3	1.3 ± 0.15
HD 198784	B2 V	4.2 ± 0.5	0.6 ± 0.2
HD 193237 ^a	B2p (LBV)	-5.0 ± 1	-8.0 ± 1
HD 198478	B2.5 Ia	2.3 ± 0.3	1.2 ± 0.15
HD 224055	B3 Ia	3.5 ± 0.3	1.4 ± 0.1
HD 28375	B3/B5 V	5.0 ± 0.5	<0.20
HD 36371	B4 Ia	3.8 ± 0.5	1.1 ± 0.2
Cyg OB2 No. 12	B5 Ia	1.8 ± 0.3	1.1 ± 0.2
HD 198183	B5/B6 V	5.5 ± 0.5	<0.15
HD 15497	B6 Ia	3.4 ± 0.3	0.9 ± 0.2
HD 21071	B7 V	5.5 ± 0.5	<0.15
HD 183143	B7 Ia	2.5 ± 0.3	0.8 ± 0.15
HD 169990	B8 V	8.0 ± 1.0	<0.15
HD 14542	B8 Ia	3.4 ± 0.3	0.3 ± 0.1
HD 34762	B9 V	8.0 ± 0.5	<0.2
HD 21291	B9 Ia	3.3 ± 0.5	0.35 ± 0.2

^a [Fe II] 1.677 μm , -2.0 ± 0.3 Å and Fe II 1.688 μm , -3.2 ± 0.5 Å.

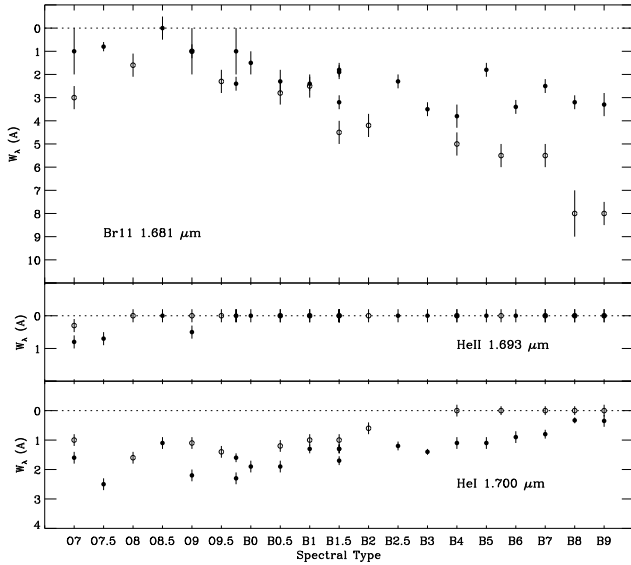


FIG. 5.—Line strength of Br 11, He II, and He I as a function of spectral type. Open circles represent dwarf stars and filled circles represent supergiants: 1σ error bars for the equivalent widths are shown as vertical bars. As is seen in the Br γ feature (HCR), there are large differences in the strength of the Br 11 feature with luminosity class within the same spectral class of late B stars. He II is only detected in O stars, and preferentially in the supergiants. Line measurements taken from P Cygni have been omitted from this figure.

of their close spectral proximity. Finally, the three *H*-band classification lines have similar intrinsic strengths in the O and early B stars (after mid B, the neutral helium lines are gone in dwarf stars). With the exception of the hydrogen lines seen in the late B dwarf stars, most features come in at just a few angstroms equivalent width. However, in order to firmly establish the existence of these lines and to study their profiles, spectra with fairly high S/N (>100) and resolution (≥ 2000) will be needed.

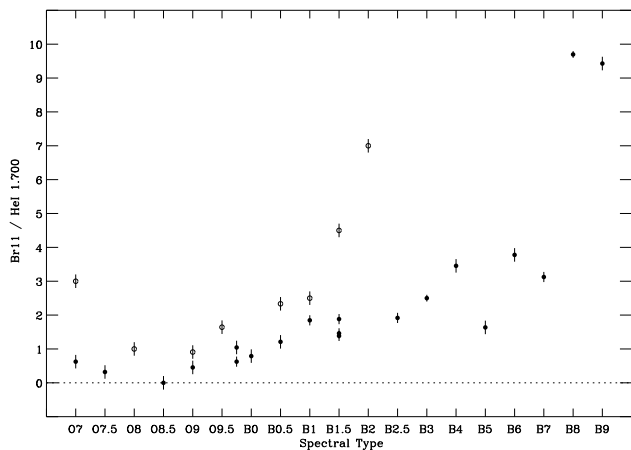


FIG. 6.—Ratio of Br 11 to 1.700 μ m He I, as seen in O and B stars. Dwarfs and supergiants are marked with open and filled circles, respectively: 1σ error bars are shown as vertical bars. This ratio is strongest among the dwarfs, as compared with the supergiants within the same spectral class. Among the late O and early B stars, where this ratio shows little change, one can still distinguish late O stars from early B stars by the presence of He II. The Br 11 to He I ratio should be very useful in estimating the luminosity class if the spectral type is previously known. The most luminous star in our sample, the B5 Ia star Cyg OB2 No. 12, shows a very weak value for this ratio. The line ratio measured in P Cygni is not included in this figure.

3.1. Spectral Classification of OB Stars in the *H* Band

Blum et al. (1997) presented a classification system for OB stars in the *H* band based on nine stars. Our more extensive sample suggests that classification should be possible, though high-quality spectra are required. We suggest the following means for identifying dwarf OB stars over the range 1.66 to 1.72 μ m using Br 11, He II, and He I. Among dwarf stars, if all three lines are present, the star is late O or earlier. Dwarf stars showing weak Br 11 ($<2\text{ \AA}$), moderately strong He I ($>1\text{ \AA}$), and no He II are from O9 to about B1. Dwarf stars showing moderately strong Br 11 (from 3 to 5 $\text{\AA}$) with a detectable He I line are mid B, B2, or B3. Dwarf stars showing strong Br 11 ($>5\text{ \AA}$) and no He I are from late B to early A. For supergiant stars, the differentiation is not as good, because of a weaker slope in the strength of Br 11 and He I with spectral type. However, we can divide the spectral characteristics into a few classes. Supergiants showing He II are late O or earlier. Supergiants with weak Br 11 ($<2\text{ \AA}$) and moderately strong He I ($>1\text{ \AA}$) are early to mid B. Supergiants with moderately strong Br 11 ($>2\text{ \AA}$) and weak He I ($<1\text{ \AA}$) are late B. Our data cannot support a more accurate classification because of the scatter in the equivalent widths seen (Fig. 5). We suggest spectral typing can be done to within about 2 or 3 spectral classes, provided the luminosity class is already known for the star.

3.2. Determining Luminosity Classifications of OB stars in the *H* Band

In our sample of stars, visual inspection reveals the dwarfs to have very broad Br 11 lines. Very broad Brackett lines can be used to identify dwarf stars if adequate resolution and S/N are obtained, such as presented here and by Meyer et al. (1998; see also Murdoch, Drew, & Anderson 1994). Another promising luminosity effect, first seen by Blum et al. (1997) in their *H*-band studies of OB stars, is the linear Stark effect (Struve 1929). The effect is to create stronger lines in the upper hydrogen series with increased luminosity of the star. We cannot see this effect in our spectra because we have only the Br 11 line in our spectral range. However, it is quite clearly demonstrated in the *H*-band spectra presented in Blum et al. (1997) and Meyer et al. (1998).

A possible new luminosity indicator is shown in Figure 6. For early and mid B stars of known spectral class, the ratio of Br 11 to He I is consistently stronger in the dwarf stars than the supergiant stars of the same spectral class. It might also be true for late O stars, though our sample size is not large enough to make such a claim fully. However, we suggest that the ratio of Br 11 to He I may prove to be a useful diagnostic of the luminosity class for B stars of known spectral type.

4. CONCLUSIONS

We have completed a study using high-S/N *H*-band spectra at a resolution of 2000 to examine the behavior of several strong lines seen in late O and B stars. Our results parallel those of Blum et al. (1997), who showed a smoothly varying change in line strengths with spectral type. However, our larger sample has revealed a luminosity effect in the strength of Br 11 and the 1.700 μ m He I line. We find that supergiant B stars show a weaker Br 11 feature and a stronger 1.700 μ m He I line compared with dwarf B stars of the same spectral class. A crude spectral typing can be

accomplished using the very small *H*-band range from 1.66 to 1.72 μm in OB stars, provided the luminosity type is known. The strength of Br 11 to He I, the strength of the higher Brackett series lines (requiring larger spectral coverage than we present in this paper), or the existence of Stark-broadened hydrogen lines, can all be used to estimate the luminosity of OB stars, though good resolution ($R \geq 2000$) and high-S/N (>100) spectra are required. Finally, the 1.700 μm He I line strengths measured as a function of spectral type and luminosity class in our survey of 34 OB stars are markedly weaker than the most recent study of this line done at one-quarter the spectral resolution by Blum et al. (1997). We stress that, when performing spectral classification, one should always observe numerous comparison spectral standard stars at similar resolution to the target stars.

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